



Lecture Slides for

INTRODUCTION TO
Machine Learning

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CHAPTER 2:

Supervised Learning

Learning a Class from Examples

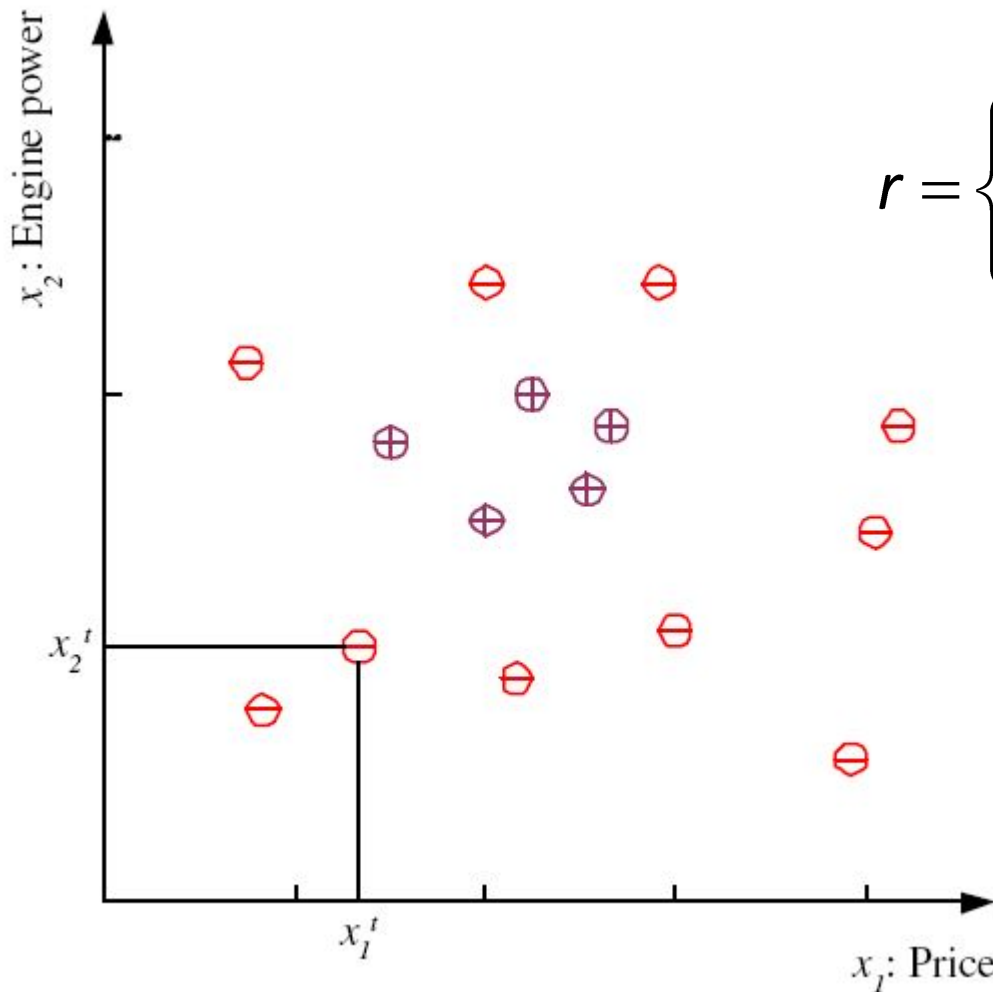
- Class C of a “family car”
 - Prediction: Is car x a family car?
 - Knowledge extraction: What do people expect from a family car?
- Output:
 - Positive (+) and negative (−) examples
- Input representation:
 - x_1 : price, x_2 : engine power

Training set X

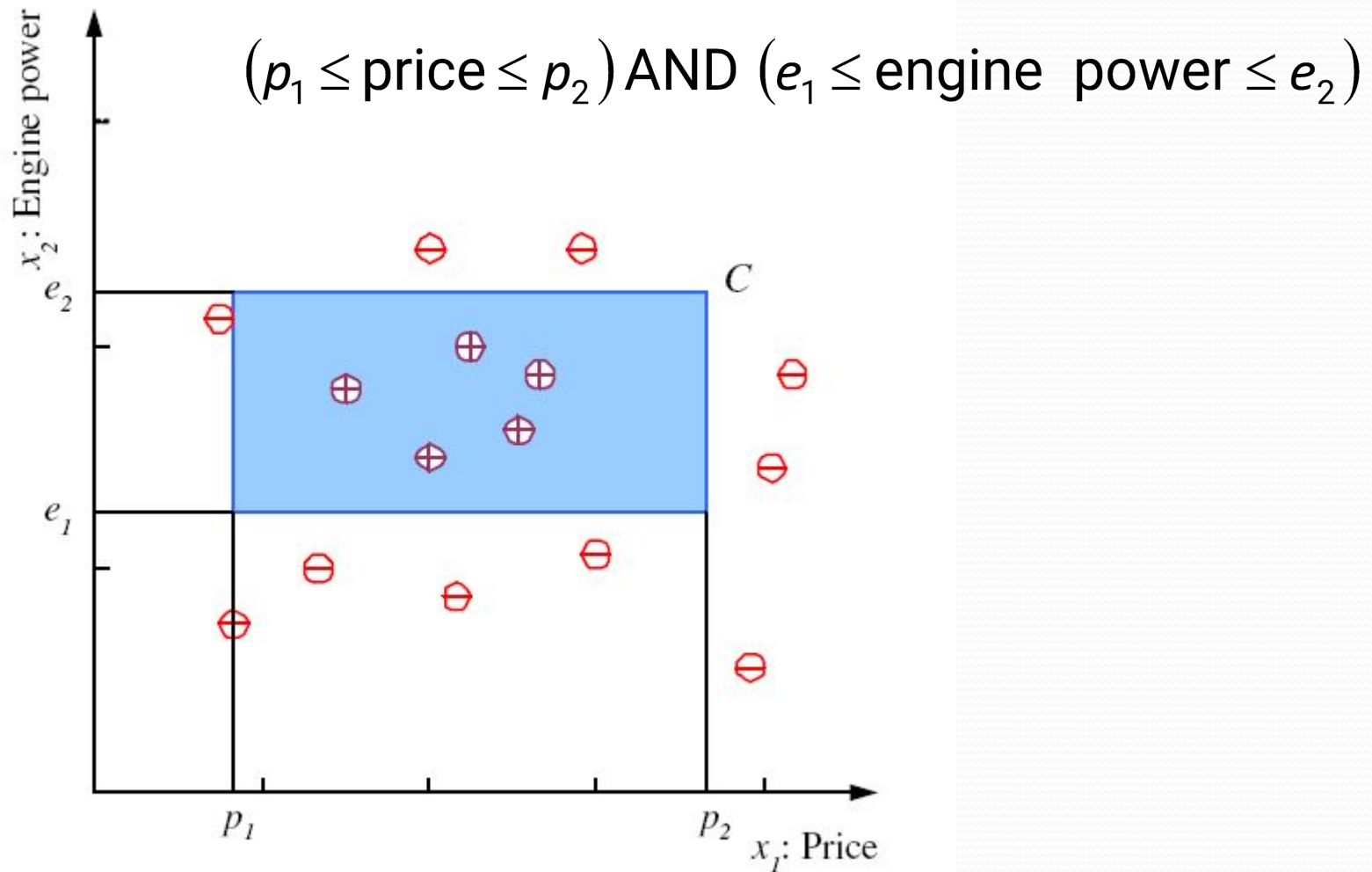
$$X = \{\mathbf{x}^t, r^t\}_{t=1}^N$$

$$r = \begin{cases} 1 & \text{if } \mathbf{x} \text{ is positive} \\ 0 & \text{if } \mathbf{x} \text{ is negative} \end{cases}$$

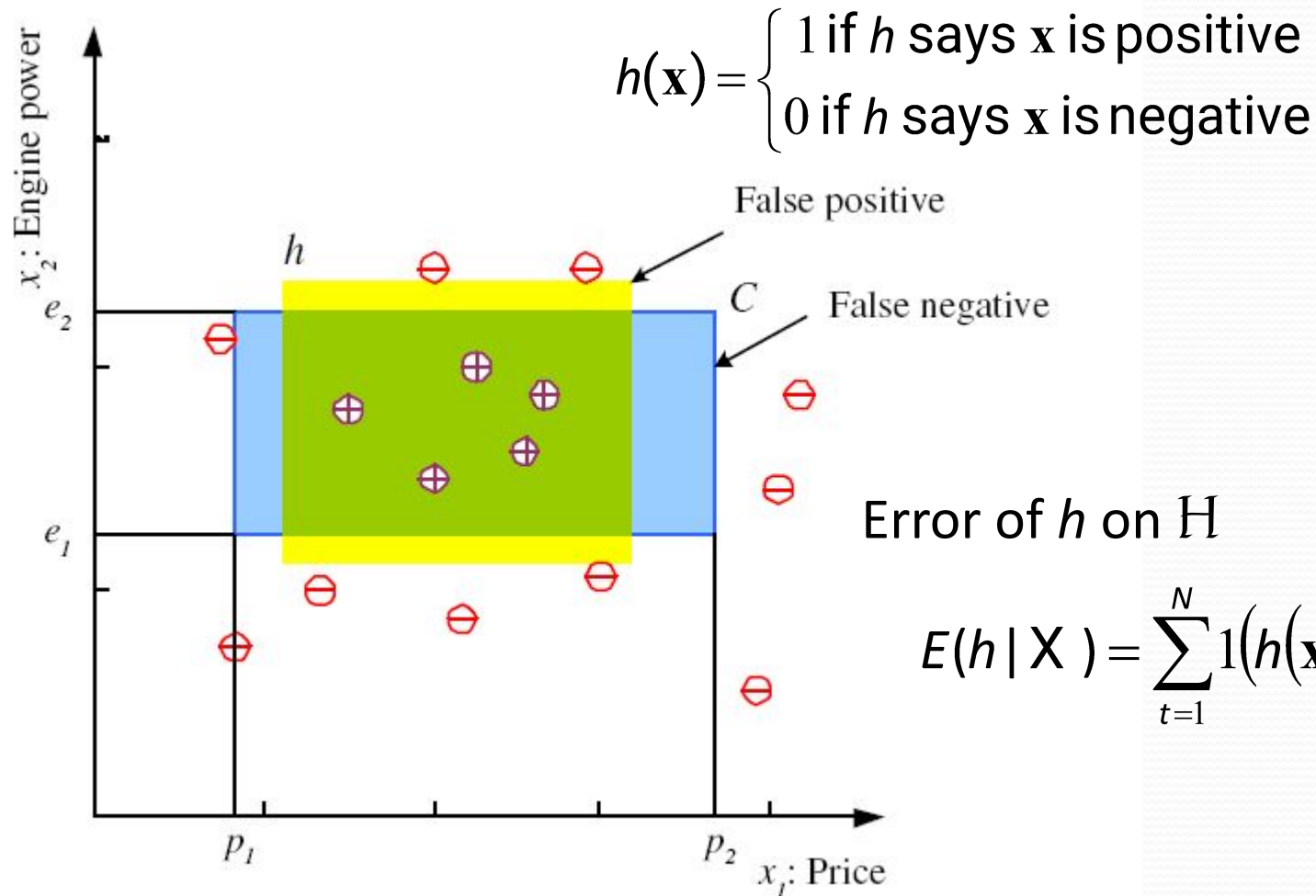
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



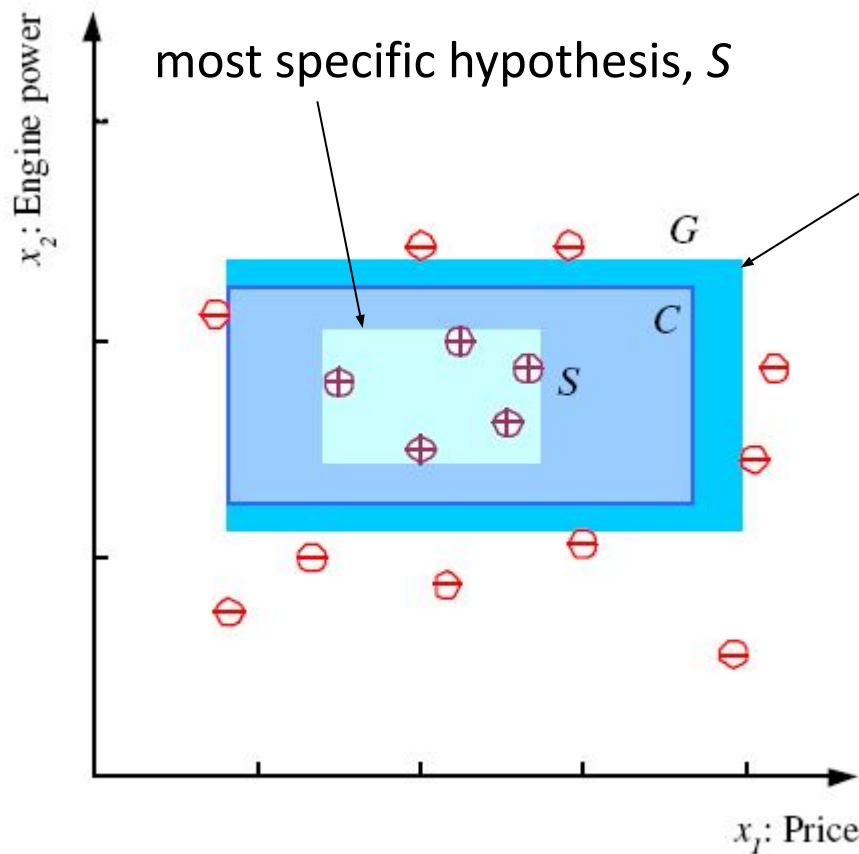
Class C



Hypothesis class $\mathcal{H}$



S, G, and the Version Space

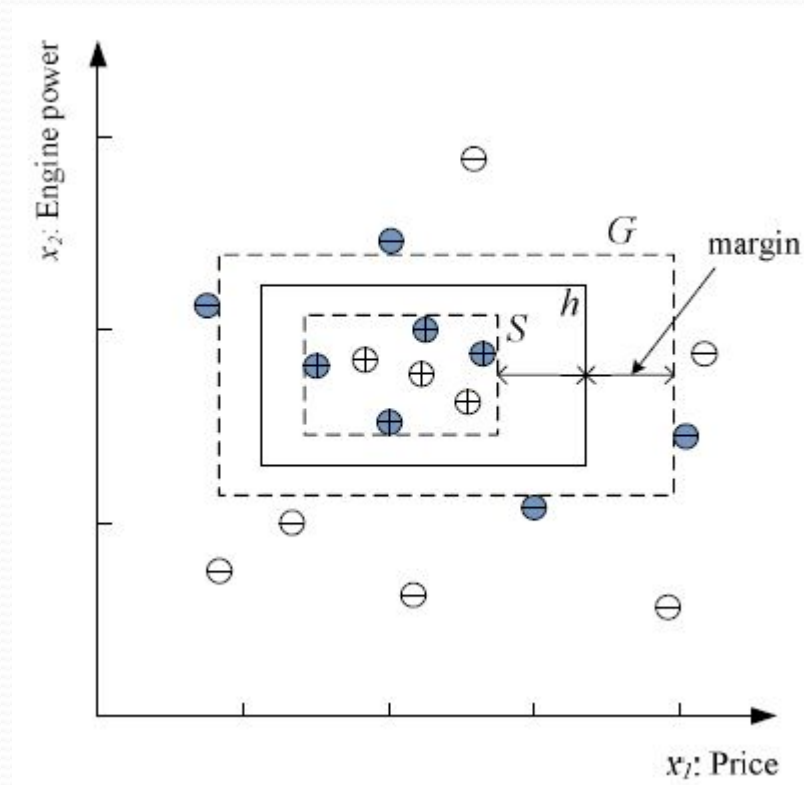


most general hypothesis, G

$h \in H$, between S and G is
consistent
and make up the
version space
(Mitchell, 1997)

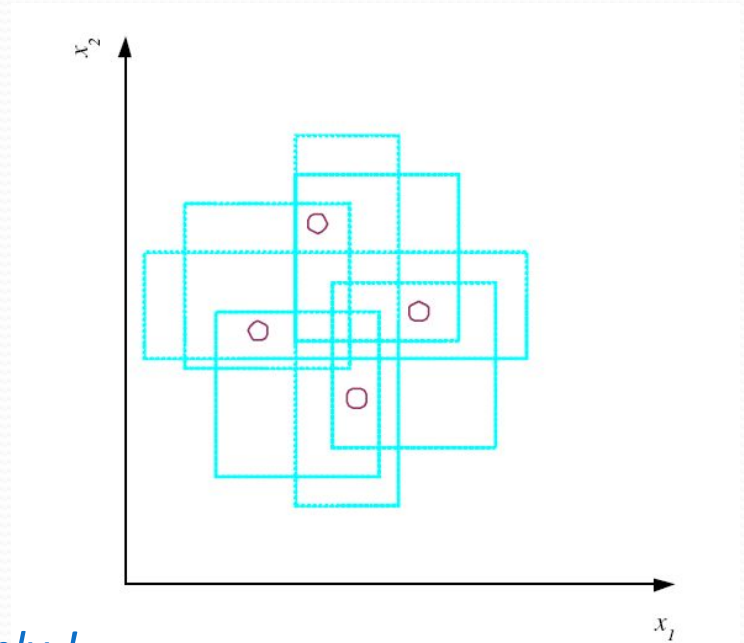
Margin

- Choose h with largest margin



VC Dimension

- N points can be labeled in 2^N ways as $+/-$
- $\mathcal{H}$ shatters N if there exists $h \in \mathcal{H}$ consistent for any of these:
$$VC(\mathcal{H}) = N$$



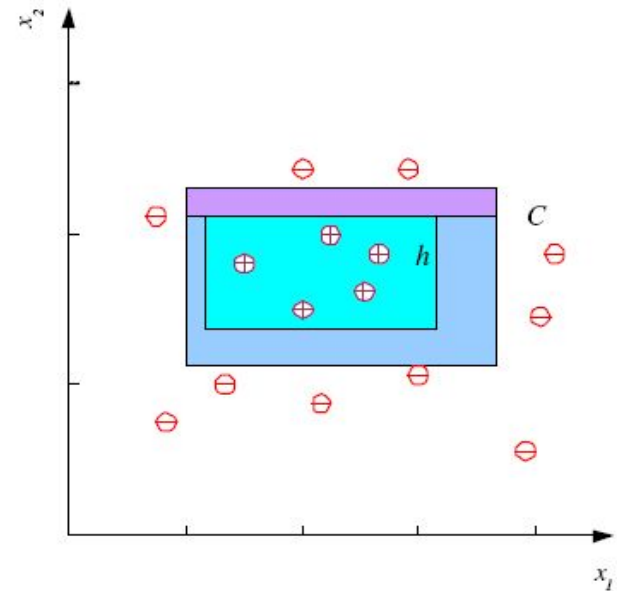
An axis-aligned rectangle shatters 4 points only !

Probably Approximately Correct (PAC) Learning

- How many training examples N should we have, such that with probability at least $1 - \delta$, h has error at most ϵ ?

(Blumer et al., 1989)

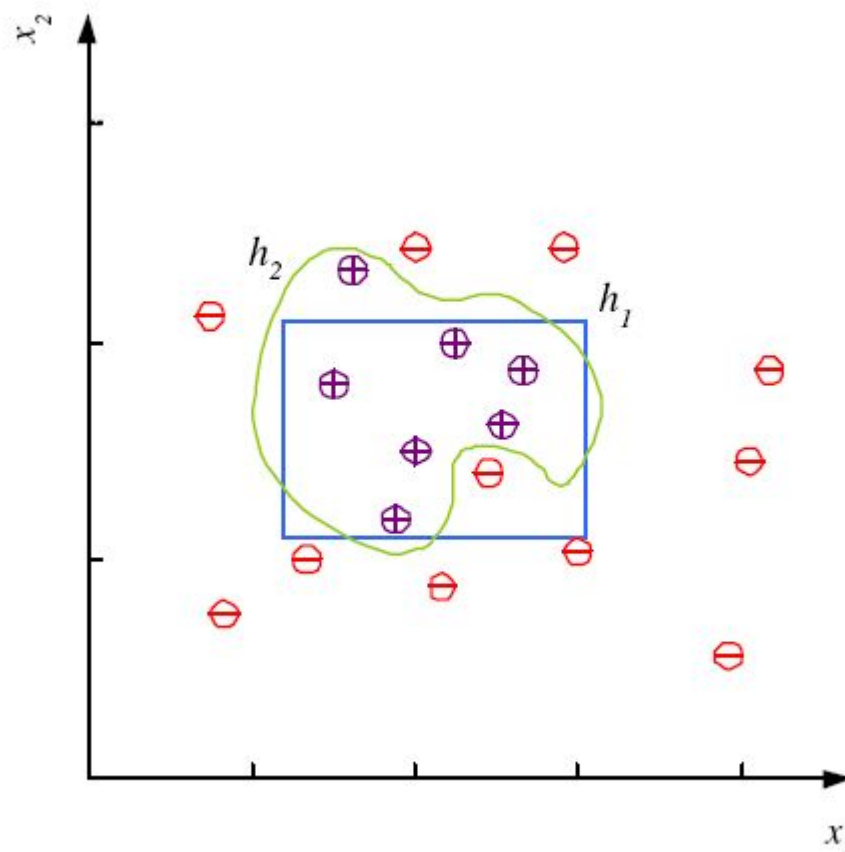
- Each strip is at most $\epsilon/4$
- Pr that we miss a strip $1 - \epsilon/4$
- Pr that N instances miss a strip $(1 - \epsilon/4)^N$
- Pr that N instances miss 4 strips $4(1 - \epsilon/4)^N$
- $4(1 - \epsilon/4)^N \leq \delta$ and $(1 - x) \leq \exp(-x)$
- $4\exp(-\epsilon N/4) \leq \delta$ and $N \geq (4/\epsilon)\log(4/\delta)$



Noise and Model Complexity

Use the simpler one because

- Simpler to use
(lower computational complexity)
- Easier to train (lower space complexity)
- Easier to explain
(more interpretable)
- Generalizes better (lower variance - Occam's razor)



Multiple Classes, C_i $i=1,\dots,K$

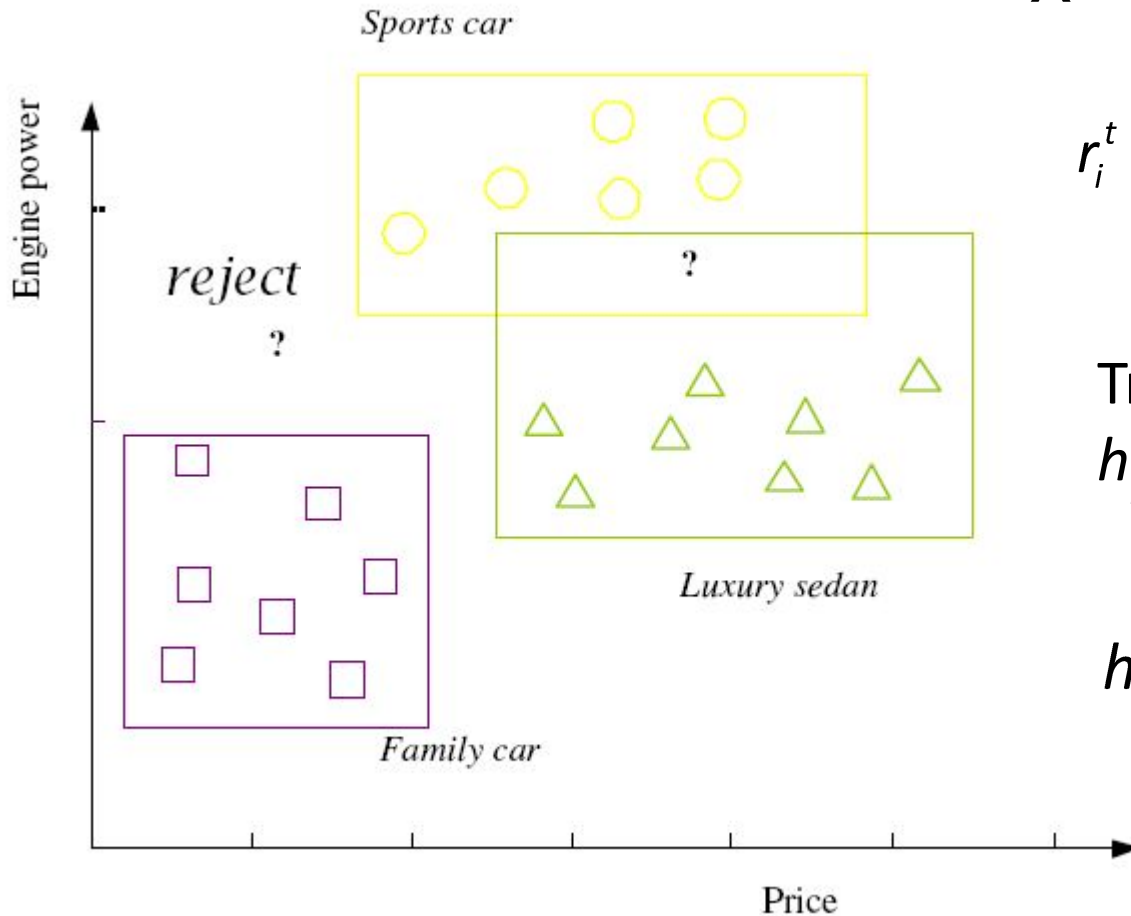
$$X = \{\mathbf{x}^t, r^t\}_{t=1}^N$$

$$r_i^t = \begin{cases} 1 & \text{if } \mathbf{x}^t \in C_i \\ 0 & \text{if } \mathbf{x}^t \in C_j, j \neq i \end{cases}$$

Train hypotheses

$h_i(\mathbf{x})$, $i=1,\dots,K$:

$$h_i(\mathbf{x}^t) = \begin{cases} 1 & \text{if } \mathbf{x}^t \in C_i \\ 0 & \text{if } \mathbf{x}^t \in C_j, j \neq i \end{cases}$$



Regression

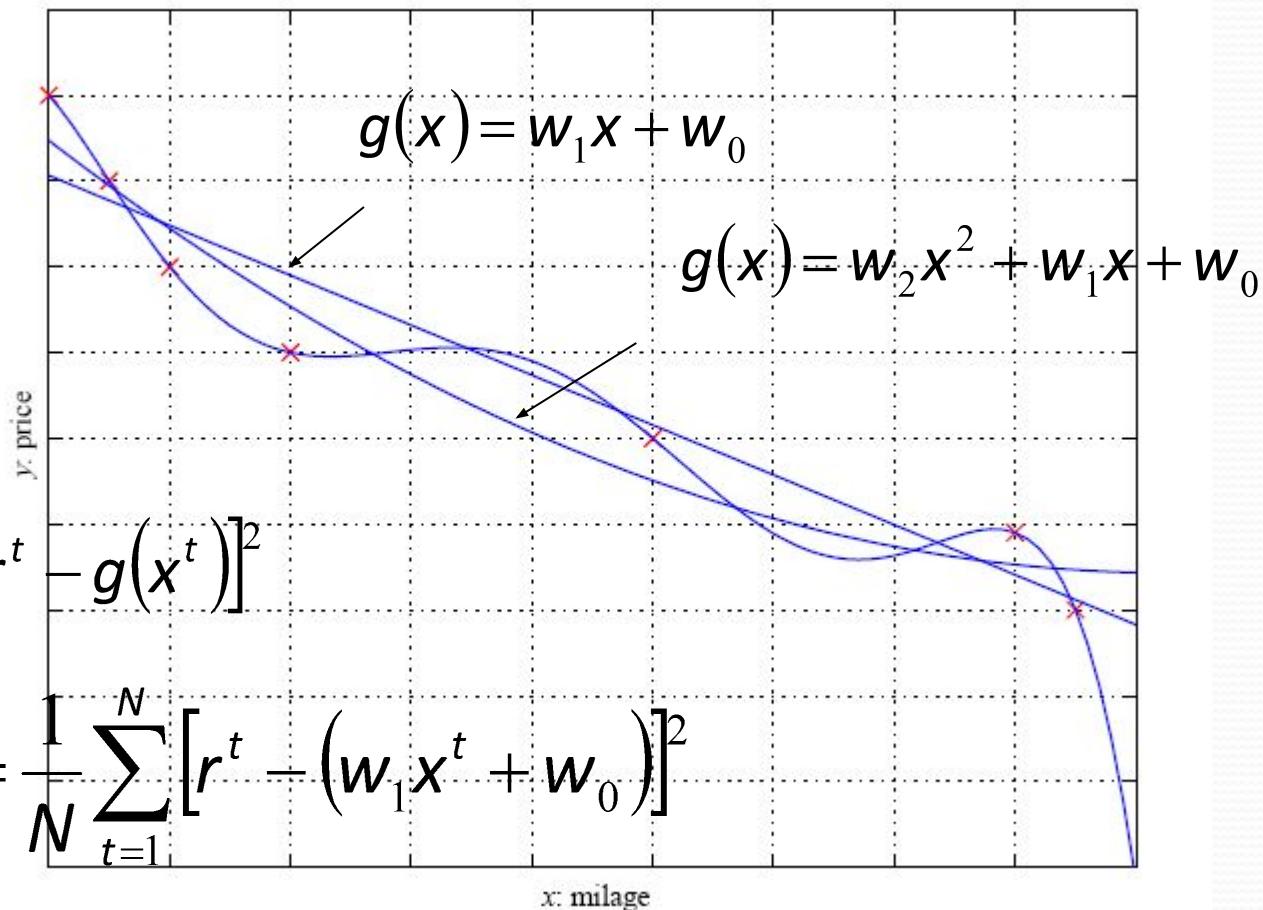
$$\mathbf{X} = \{x^t, r^t\}_{t=1}^N$$

$$r^t \in \mathbb{R}$$

$$r^t = f(x^t) + \varepsilon$$

$$E(g | \mathbf{X}) = \frac{1}{N} \sum_{t=1}^N [r^t - g(x^t)]^2$$

$$E(w_1, w_0 | \mathbf{X}) = \frac{1}{N} \sum_{t=1}^N [r^t - (w_1 x^t + w_0)]^2$$



Model Selection & Generalization

- Learning is an ill-posed problem; data is not sufficient to find a unique solution
- The need for inductive bias, assumptions about H
- Generalization: How well a model performs on new data
- Overfitting: H more complex than C or f
- Underfitting: H less complex than C or f

Triple Trade-Off

- There is a trade-off between three factors (Dietterich, 2003):
 1. Complexity of H , $c(H)$,
 2. Training set size, N ,
 3. Generalization error, E , on new data
- As $N \uparrow$, $E \downarrow$
- As $c(H) \uparrow$, first $E \downarrow$ and then $E \uparrow$

Cross-Validation

- To estimate generalization error, we need data unseen during training. We split the data as
 - Training set (50%)
 - Validation set (25%)
 - Test (publication) set (25%)
- Resampling when there is few data

Dimensions of a Supervised Learner

1. Model: $g(\mathbf{x} | \theta)$

2. Loss function: $E(\theta | X) = \sum_t L(r^t, g(\mathbf{x}^t | \theta))$

3. Optimization procedure:

$$\theta^* = \arg \min_{\theta} E(\theta | X)$$